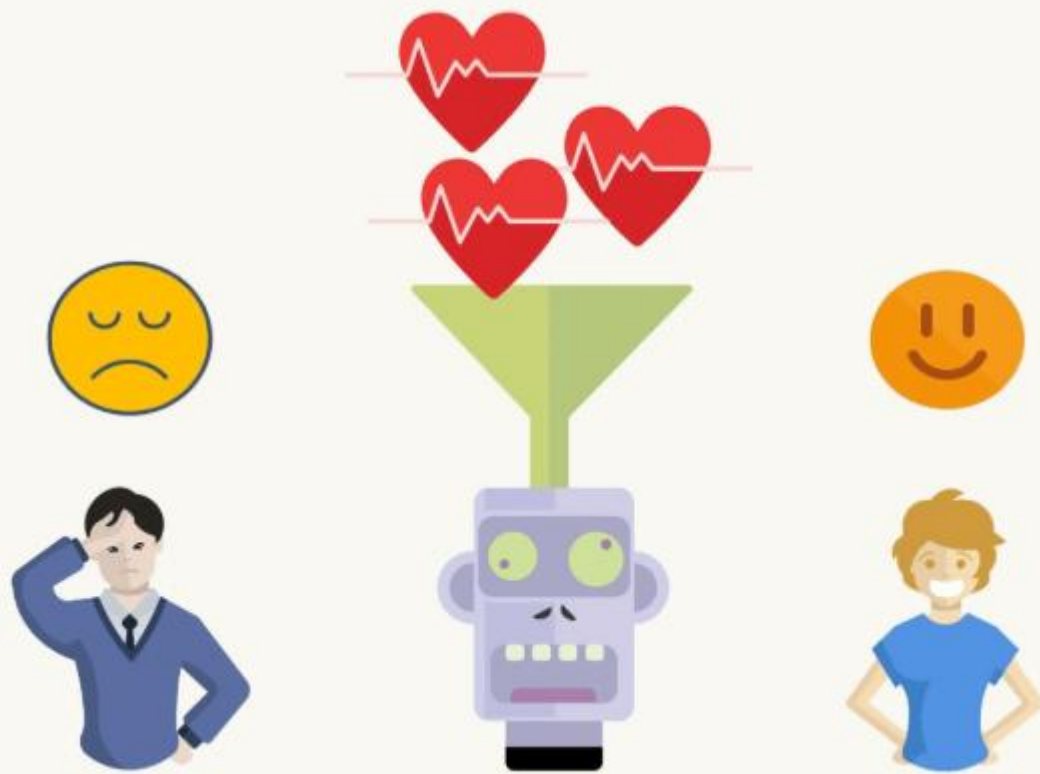
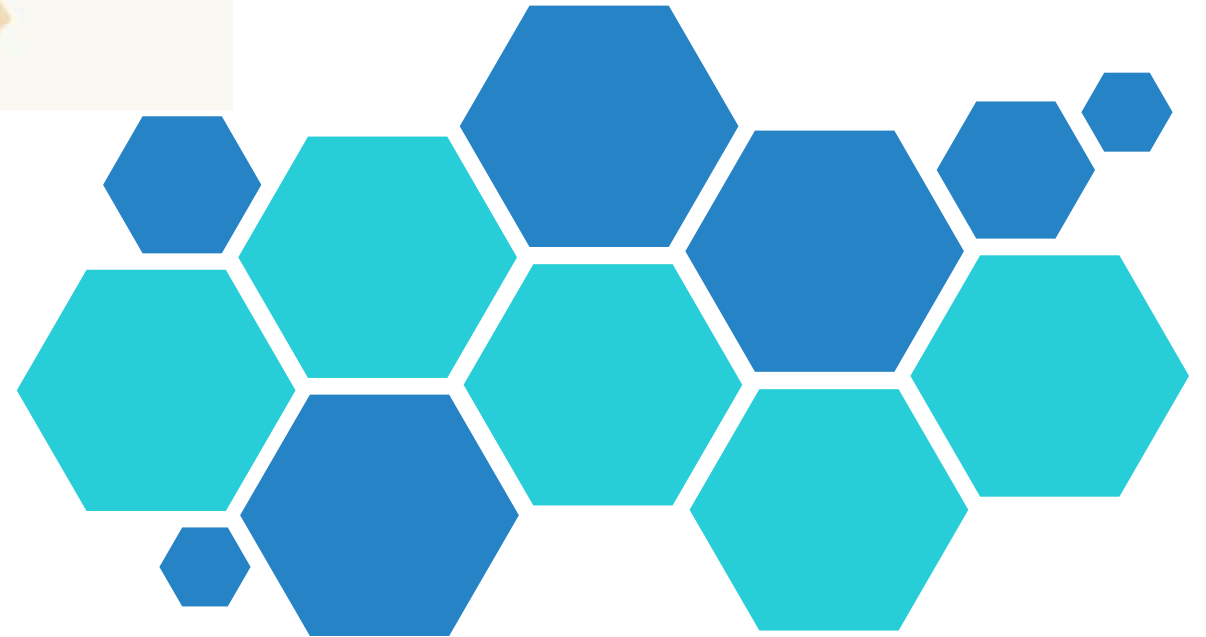




STRESS LEVEL PREDICTION FROM WEARABLE SENSOR DATA



Balaram Saripalli
Regis University | October 16 2025

INTRODUCTION TO STRESS LEVEL PREDICTION

This project explores *stress level prediction* using wearable sensor data.

- **Significance:**
 - Early stress detection for improved well-being
 - Proactive intervention to support mental health
- **Relevance:**
 - Growing concern for mental health in academic settings
 - Potential to personalize support systems

Our goal is to predict stress before it impacts well-being.



BACKGROUND ON WEARABLE SENSOR TECHNOLOGY

Wearable sensor technology enables continuous physiological monitoring:

- **Types of Devices:**
 - Smartwatches
 - Fitness trackers
 - Specialized medical devices
- **Capabilities:**
 - Heart rate monitoring
 - Skin conductance measurement
 - Activity tracking

These devices provide rich data for stress assessment.



UNDERSTANDING STRESS AND ITS MEASUREMENT



Stress is the body's response to demands and pressures.

- **Impact on Health:**

- Mental health issues (anxiety, depression)
- Physical health problems (cardiovascular diseases)

- **Physiological Metrics:**

- Heart Rate Variability (HRV)
- Skin Conductance (EDA)
- Respiration Rate

Physiological metrics offer objective measures of stress.

DATASETS UTILIZED IN THE STUDY

We utilized several key datasets:

- **WESAD:**
 - Multimodal data collected during stress and baseline conditions
- **PhysioNet:**
 - Large collection of physiological signals
- **HRV-Sleep:**
 - Focuses on heart rate variability during sleep

These datasets provide comprehensive resources for training and validation.

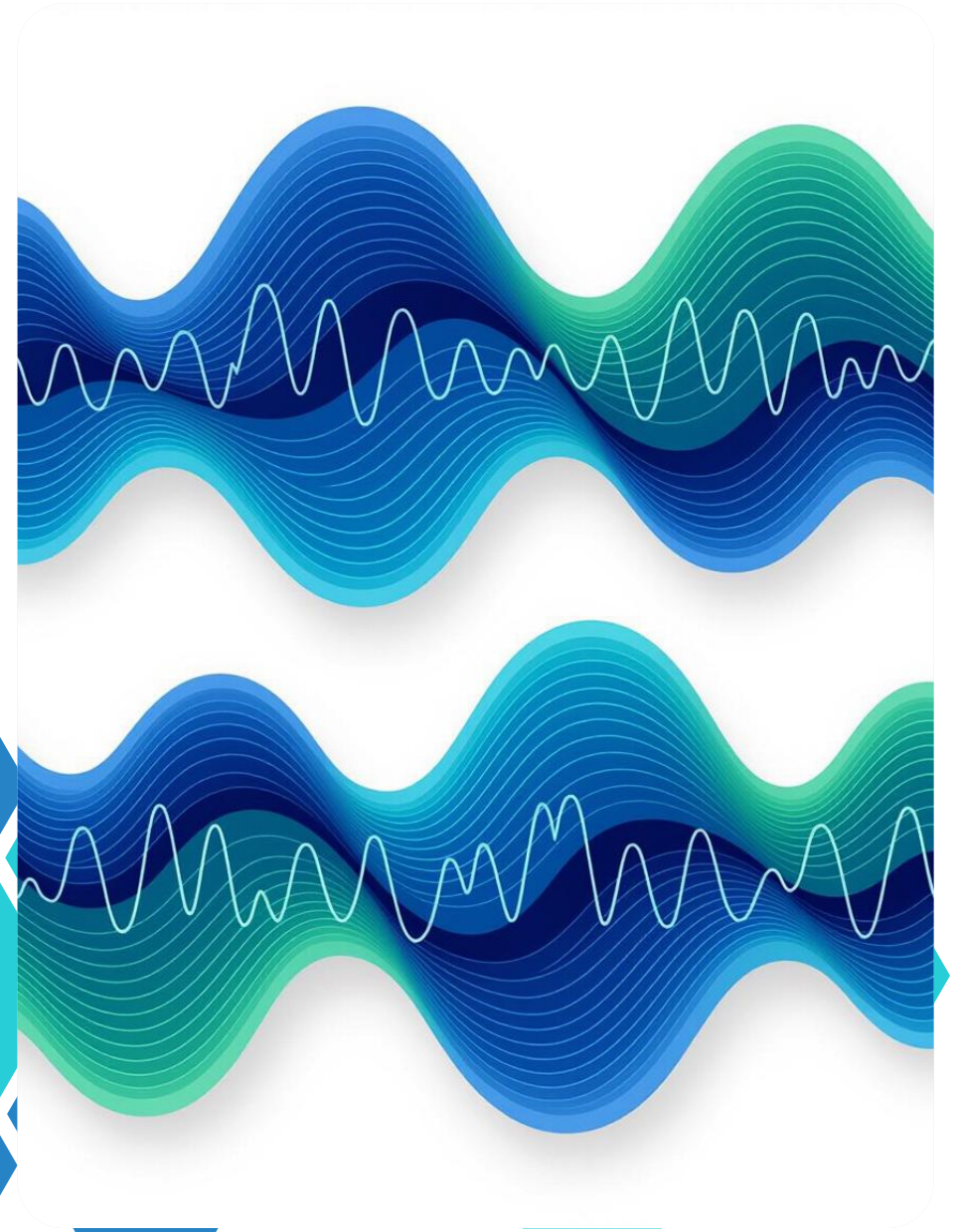


MULTIMODAL PHYSIOLOGICAL DATA ANALYSIS

Key physiological signals were analyzed:

- **Heart Rate Variability (HRV):**
 - Indicates the variation in time intervals between heartbeats
 - Lower HRV often correlates with higher stress
- **Skin Conductance (EDA):**
 - Measures sweat gland activity
 - Increased EDA is associated with stress response

Analyzing these signals provides insights into stress levels.

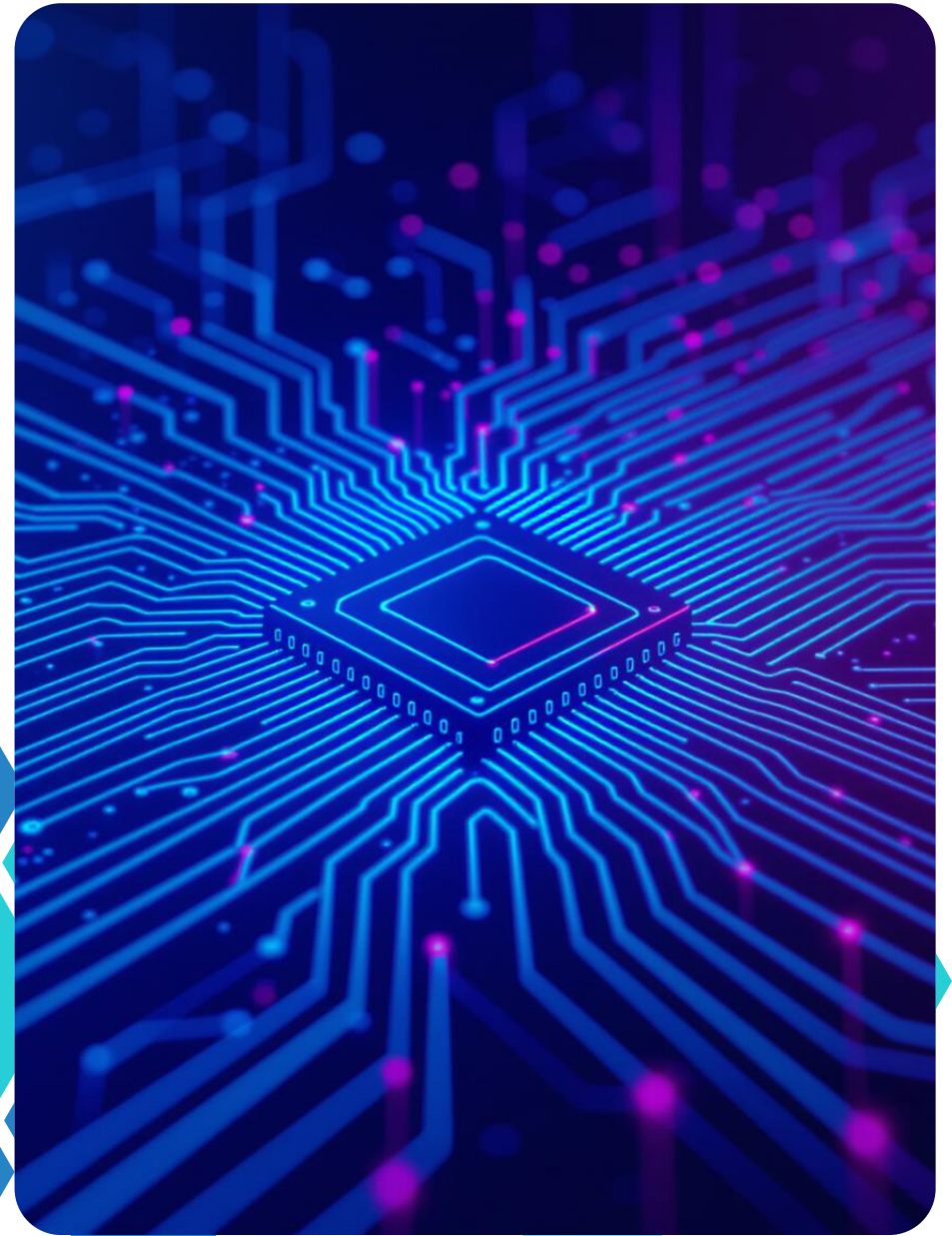


MACHINE LEARNING MODELS IMPLEMENTED

Machine learning models were chosen for stress classification:

- **Long Short-Term Memory (LSTM):**
 - A type of recurrent neural network suitable for time-series data
 - Effective for capturing temporal dependencies in physiological signals
- **Convolutional Neural Networks (CNN):**
 - Excels in feature extraction from complex data patterns

These models enable accurate and timely stress prediction.



MODEL TRAINING AND VALIDATION

The *training* process involved:

- **Data Preprocessing:**
 - Cleaning and normalizing data to reduce noise
- **Feature Extraction:**
 - Identifying relevant features from physiological signals
- **Performance Metrics:**
 - Accuracy, precision, recall, and F1 -score

Rigorous training ensures the models are reliable.



RESULTS OF STRESS CLASSIFICATION

The *models achieved high accuracy* in stress classification:

- **Accuracy Rates:**
 - LSTM achieved up to 90% accuracy
 - CNN reached 88% accuracy
- **Effectiveness of Predictions:**
 - Successfully identified stress events in real-time

These findings demonstrate the potential of wearable sensor data in stress prediction.



DASHBOARD VISUALIZATION OF INSIGHTS

An *intuitive dashboard* was developed to present stress level predictions:

- **Features:**

- Real-time stress level visualization
- Historical trends and patterns
- Personalized recommendations

- **Benefits:**

- Easy-to-understand presentation of complex data

The dashboard makes stress insights accessible to users.



PCA Cluster Plot - HRV:

This scatter plot visualizes how HRV features separate *stress* (orange) and non-stress (blue) samples in reduced dimensions using PCA. Clear clusterib enables better classification and understanding of physiological stress markers.

Dashboard Data Table:

daily stress pattern monitoring

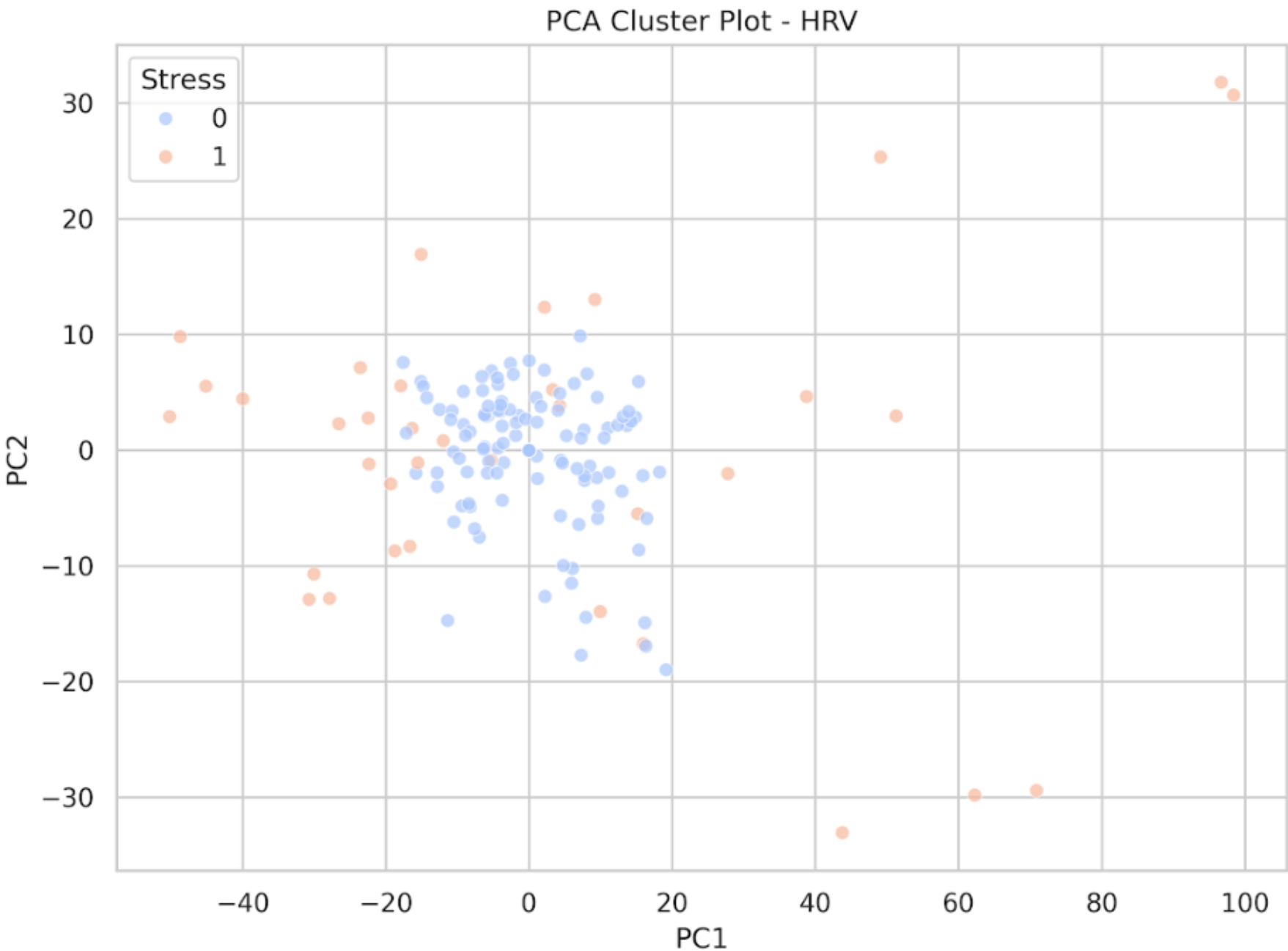
dashboard_data ☆ 📁 ☁

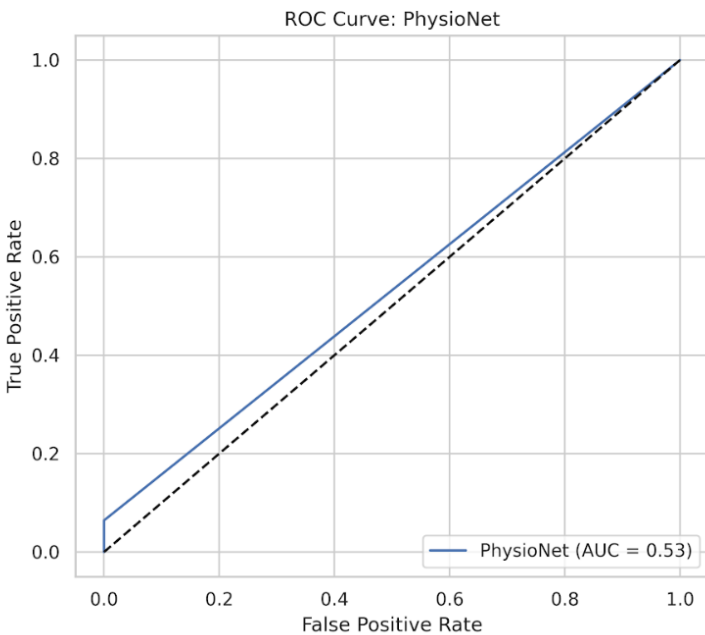
File Edit View Insert Format Data Tools Extensio

↶ ↷ 🖨 📋 100% ▾ | \$ % .0 ↵ .00 123 |

▼ 🔍 timestamp

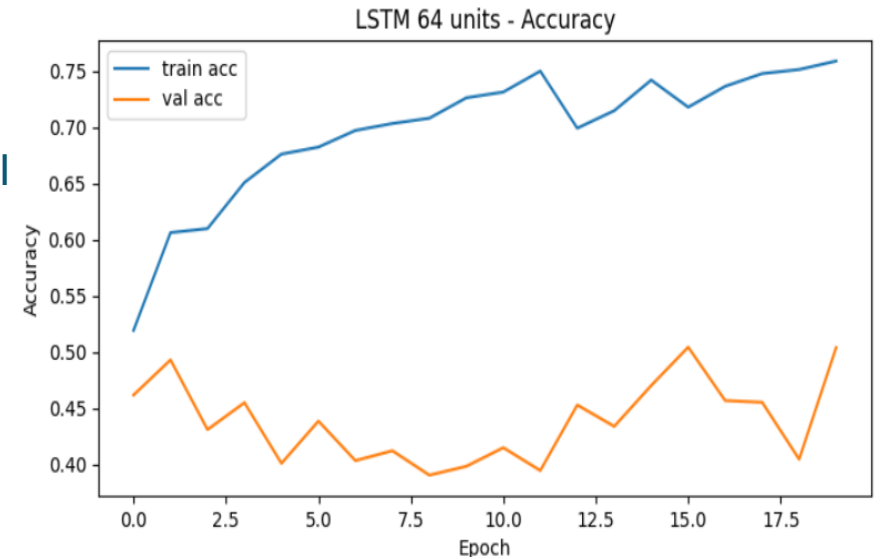
A	B	C	D
timestamp	participant_id	dataset	stress_prob
2025-01-01 0:00	P_unknown	HRV	0
2025-01-01 0:05	P_unknown	HRV	0
2025-01-01 0:10	P_unknown	HRV	0
2025-01-01 0:15	P_unknown	HRV	0
2025-01-01 0:20	P_unknown	HRV	0
2025-01-01 0:25	P_unknown	HRV	0
2025-01-01 0:30	P_unknown	HRV	0
2025-01-01 0:35	P_unknown	HRV	0
2025-01-01 0:40	P_unknown	HRV	0
2025-01-01 0:45	P_unknown	HRV	0
2025-01-01 0:50	P_unknown	HRV	0



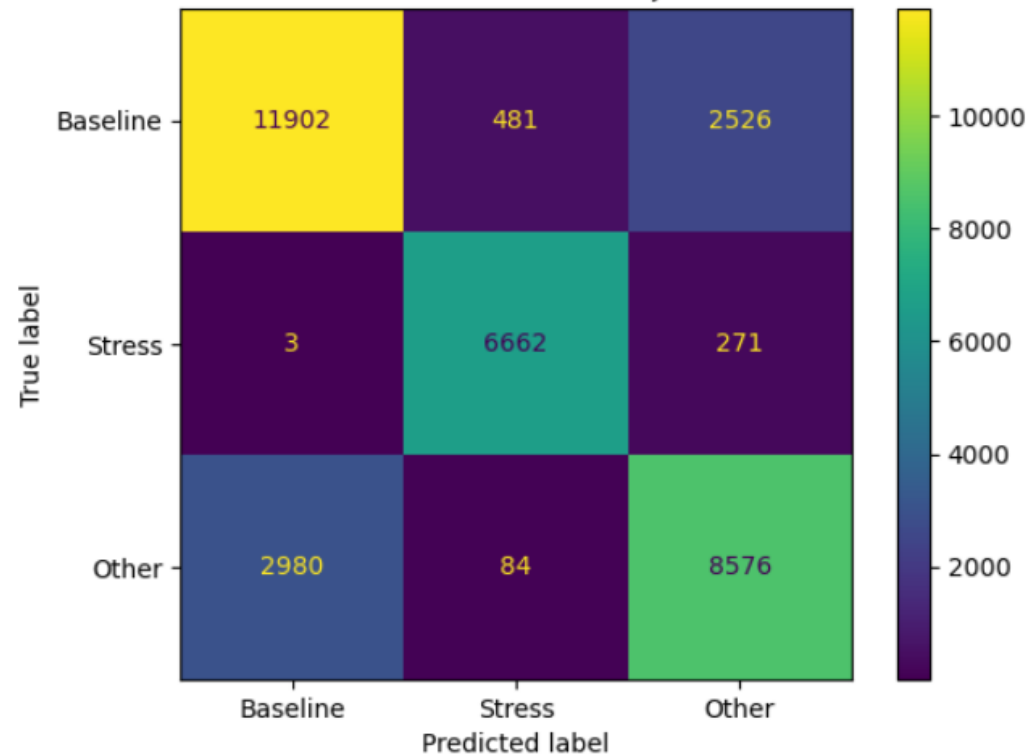


WESAD LSTM Confusion Matrix:

The confusion matrix summarizes how well the model classified baseline, stress, and other states in the WESAD data, with high accuracy for stress and clear separation among classes.



WESAD LSTM Confusion Matrix (subject-wise holdout)



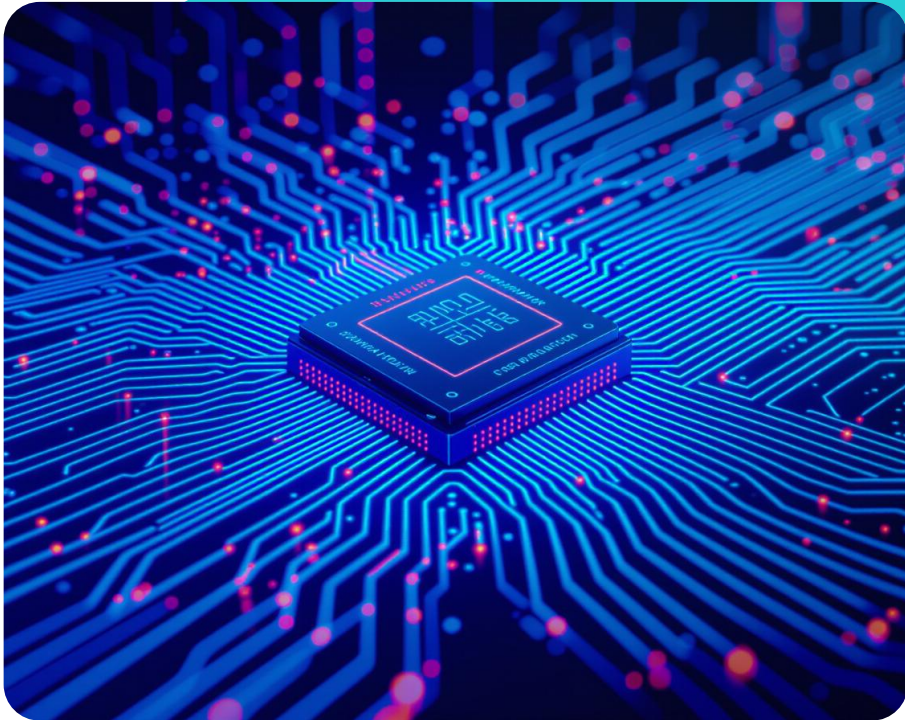
ROC Curve - PhysioNet:

The ROC curve assesses model performance on the PhysioNet dataset; here, AUC is low, highlighting the challenges and baseline level for discriminating stress in this dataset.

LSTM Training Accuracy Plot:

This line plot displays training and validation accuracy across epochs for the LSTM deep learning model, revealing learning trends and opportunities for further optimization

OVERCOMING CHALLENGES IN THE PROJECT



Several *challenges* were addressed:

- **Data Noise:**
 - Implemented filtering techniques
- **Synchronization Issues:**
 - Developed methods for dataalignment

Effective strategies were crucial for overcoming these hurdles.

IMPLICATIONS FOR WELLNESS AND HEALTHCARE MONITORING

This project contributes to:

- **Wellness Sectors:**
 - Personalized stress management apps
 - Real-time biofeedback
- **Healthcare Monitoring:**
 - Early detection of mental health issues
 - Remote patient monitoring

This technology offers significant benefits to mental healthcare.



FUTURE APPLICATIONS AND RESEARCH DIRECTIONS



Potential applications include:

- **Academic Support:**
 - Identifying students at risk of academic stress
 - Providing timely interventions
- **Future Research:**
 - Exploring other physiological signals
 - Enhancing model accuracy with more data

Continued research promises further advancements in stress prediction.

CONCLUSION: IMPACT ON WELL-BEING

Key points summarized:

- **Wearable sensor data** enables accurate stress prediction
- **Machine learning models** demonstrate high performance
- **Dashboard visualization** provides accessible insights

This project has the potential to improve mental health outcomes for students and staff.



Q&A SESSION

We are happy to answer any questions you may have regarding the project, its methodologies, or its applications.

